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Sr. No. of Question Paper : 852 E Your Roll No.....

Unique Paper Code : 234403

Name of the Course : B.Sc. (H) Computer Science

Name of the Paper : Data Communication and Computer Networks (CSHT-409)

Semester : IV

Duration : 3 Hours

Maximum Marks : 75

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. All questions in **Section-A** are compulsory.
3. Attempt **any four** questions in **Section-B**.

SECTION – A

(All questions are compulsory.)

1. (a) Explain which layer in the OSI model performs the following functions :
 - (i) Route Determination
 - (ii) Error Detection and Correction
 - (iii) Interface to outside world
 - (iv) Services such as e-mail and file transfer (4)
- (b) What is the purpose of DF and MF bits in IPV4 Header ? (2)
- (c) At what layer(s) do the following protocols operate in TCP/IP protocol ?
 - (i) FTP

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- (ii) ICMP
- (iii) DNS
- (iv) UDP (4)
- (d) How is Selective Repeat better than Go Back N ? (3)
- (e) Evaluate the maximum bit rate for a channel having bandwidth 1600 Hz if S/N ratio is 20db ? (2)
- (f) A constellation diagram consists of eight equally spaced points on a circle. If the bit rate is 4800 bps, what is the baud rate ? (3)
- (g) What is the frame format of PPP ? (2)
- (h) A receiver receives the vector 11001101111. Using the Hamming code algorithm, what was the original code sent. (5)
- (i) What are the different components of a URL ? Explain it. (3)
- (j) What is a loop back IP address ? How is it useful ? (3)
- (k) Write a short note on the following :
 - (i) Peer to peer connection
 - (ii) RPC (2×2)

SECTION –B

(Attempt any four questions from Section B)

2. (a) What is the difference between flow control and congestion control ? Also specify the reason for occurrence of congestion in a network. (7)

- (b) What is meant by CRC ? How is it useful in networks ? (3)
3. (a) What is Binary Exponential Back off algorithm in Ethernet ? How does it reduce the probability of collision in the Ethernet ? (2+4)
- (b) What is the format of IP datagram ? (4)
4. (a) State and examine the distance vector routing algorithm. Briefly discuss the Count to Infinity problem. (6)
- (b) Describe the TCP connection establishment and release procedure. (4)
5. (a) What do mean by data communication ? Write the characteristics on which effectiveness of data communication system depends. (5)
- (b) Define the following terms :
- (i) Unicasting
 - (ii) Broadcasting
 - (iii) Multicasting (3)
- (c) Write the short note on WWW. (2)
6. (a) The network 134.122.0.0/16 has been subdivided into /19 networks.
- (i) How many /19 sub networks are there. Give their addresses.
 - (ii) How many hosts can be on each network.
 - (iii) Determine which network the IP address 134,122.67.124 belongs to. (6)
- (b) Explain the features of Datalink layer and Transport layer. (4)

P.T.O.

7. (a) Discuss the thick and thin Ethernet. (3)
- (b) How does DPCM differ from PCM? What does adaptive DPCM do to achieve compression? (4)
- (c) Draw a diagram showing A, B, C, D and E IP address formats. (3)